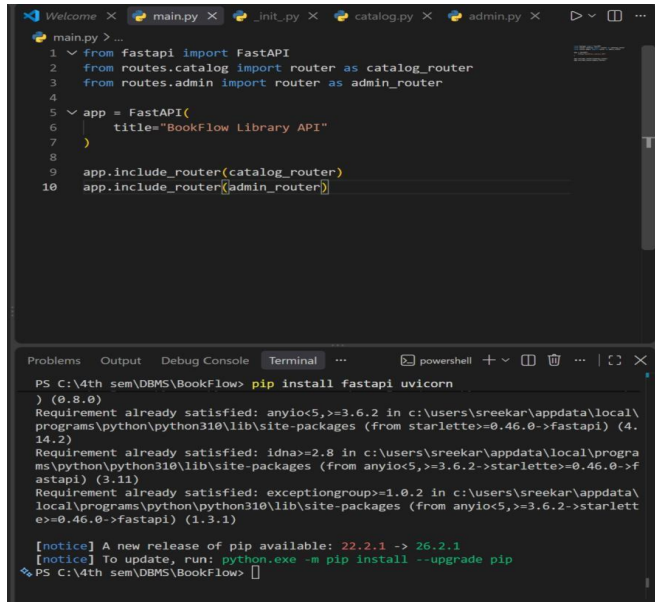


PRCATICAL 6
NAME: SATWIK
ROLL NO: 2520090151



The image shows a Visual Studio Code editor with a Python file named `main.py` and a PowerShell terminal window. The `main.py` file contains the following code:

```
1 from fastapi import FastAPI
2 from routes.catalog import router as catalog_router
3 from routes.admin import router as admin_router
4
5 app = FastAPI(
6     title="BookFlow Library API"
7 )
8
9 app.include_router(catalog_router)
10 app.include_router(admin_router)
```

The PowerShell terminal window shows the output of the command `pip install fastapi uvicorn`. The output indicates that the requirements are already satisfied for `anyio`, `idna`, and `exceptiongroup`. A notice at the bottom suggests updating `pip` to version 26.2.1.

```
PS C:\4th sem\DBMS\BookFlow> pip install fastapi uvicorn
) (0.8.0)
Requirement already satisfied: anyio<5,>=3.6.2 in c:\users\sreekar\appdata\local\
programs\python\python310\lib\site-packages (from starlette=0.46.0->fastapi) (4.
14.2)
Requirement already satisfied: idna>=2.8 in c:\users\sreekar\appdata\local\progra
ms\python\python310\lib\site-packages (from anyio<5,>=3.6.2->starlette=0.46.0->f
astapi) (3.11)
Requirement already satisfied: exceptiongroup>=1.0.2 in c:\users\sreekar\appdata\
local\programs\python\python310\lib\site-packages (from anyio<5,>=3.6.2->starlett
e=0.46.0->fastapi) (1.3.1)

[notice] A new release of pip available: 22.2.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\4th sem\DBMS\BookFlow>
```

The screenshot shows the Visual Studio Code editor with the 'catalog.py' file open. The file contains the following code:

```
routes > catalog.py > get_book
20     "status": "Available"
21 }
22 ]
23
24 @router.get("/books")
25 def get_books():
26     return books
27
28 @router.get("/books/{book_id}")
29 def get_book(book_id: int):
30     for book in books:
31         if book["id"] == book_id:
32             return book
33
34     raise HTTPException(
35         status_code=404,
36         detail="Book not found"
37     )
```

The terminal window shows the output of the command `pip install fastapi uvicorn`:

```
PS C:\4th sem\DBMS\BookFlow> pip install fastapi uvicorn
fastapi (0.8.0)
Requirement already satisfied: anyio<5,>=3.6.2 in c:\users\sreekar\appdata\loc
al\programs\python\python310\lib\site-packages (from starlette=>0.46.0->fastap
i) (4.14.2)
Requirement already satisfied: idna>=2.8 in c:\users\sreekar\appdata\local\pro
grams\python\python310\lib\site-packages (from anyio<5,>=3.6.2->starlette=>0.4
6.0->fastapi) (3.11)
Requirement already satisfied: exceptiongroup>=1.0.2 in c:\users\sreekar\appda
ta\local\programs\python\python310\lib\site-packages (from anyio<5,>=3.6.2->st
arlette=>0.46.0->fastapi) (1.3.1)

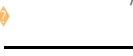
[notice] A new release of pip available: 22.2.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\4th sem\DBMS\BookFlow>
```

The terminal window shows the output of the command `uvicorn main:app --reload`:

```
PS C:\4th sem\DBMS\BookFlow> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\\4th sem\\DBMS\\BookF
low']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [28444] using StatReload
INFO: Started server process [35296]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

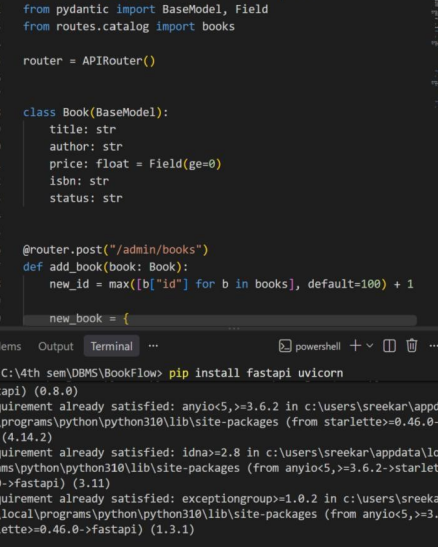
BookFlow Library API

loopenapqson

default		A
	/books Get Books	
	/books/{book_id} Get Book	V
	/admin/books Add Book	V
	/admin/books/{book_id} Update Book	V
	/admin/books/{book_id} Delete Book	V

Schemas

Book	>	pn	object
HTTPValidationError	>		object
ValidationError	>		object



The image shows a VS Code editor with a file named `delete_book.py` open. The code defines a `Book` model using `BaseModel` and a `delete_book` endpoint using `APIRouter`. The terminal window shows the command `pip install fastapi uvicorn` being executed, with output indicating that the packages are already satisfied or successfully installed.

```

Welcome | main.py | _init.py | catalog.py | admin.py |
routes > admin.py > delete_book
1 from fastapi import APIRouter, HTTPException
2 from pydantic import BaseModel, Field
3 from routes.catalog import books
4
5 router = APIRouter()
6
7
8 class Book(BaseModel):
9     title: str
10    author: str
11    price: float = Field(ge=0)
12    isbn: str
13    status: str
14
15
16 @router.post("/admin/books")
17 def add_book(Book: Book):
18     new_id = max([b["id"] for b in books], default=100) + 1
19
20     new_book = {

```

```

Problems Output Terminal ... powershell + - - - - -
PS C:\4th sem\DBMS\BookFlow> pip install fastapi uvicorn
astapi) (0.8.0)
Requirement already satisfied: anyio<5,>=3.6.2 in c:\users\sreekar\appdata\loc
al\programs\python\python310\lib\site-packages (from starlette=>0.46.0->fastap
i) (4.14.2)
Requirement already satisfied: idna>=2.8 in c:\users\sreekar\appdata\local\pro
grams\python\python310\lib\site-packages (from anyio<5,>=3.6.2->starlette=>0.4
6.0->fastapi) (3.11)
Requirement already satisfied: exceptiongroup>=1.0.2 in c:\users\sreekar\appda
ta\local\programs\python\python310\lib\site-packages (from anyio<5,>=3.6.2->st
arlette=>0.46.0->fastapi) (1.3.1)

[notice] A new release of pip available: 22.2.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\4th sem\DBMS\BookFlow>

```

Request body required

application/json

Edit Value Schema

```
"title": "Machine Learning Basics",
"author": "Alice Johnson",
"price": 600,
"isbn": "9781111111111",
"status": "Available"
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/admin/books' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Machine Learning Basics",
    "author": "Alice Johnson",
    "price": 600,
    "isbn": "9781111111111",
    "status": "Available"
  }'
```

book_id • reqwtd

integer
(path)

101

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/books/101' \
  -H 'accept: */*' |
```



Request URL

http://127.0.0.1:8000/books/101

Server response

Code Details

200

Response body

```
{
  "id": 101,
  "title": "Python Progrmm1ng",
  "author": "John Smith",
  "price": 100,
  "isbn": "9781234567890",
  "status": "Avulabh"
}
```



Download

Response headers

```
content-length: 117
content-type: application/json
date: Sun, 20 Sep 2026 18:32:16 GMT
server: uvicorn
```

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/admin/books' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Machine Learning Basics",
    "author": "Alice Johnson",
    "price": 600,
    "isbn": "9781111111111",
    "status": "Available"
  }'
```

Request URL

http://127.0.0.1:8000/admin/books

Server response

Code Details

200 Response body

```
{
  "message": "Book added successfully",
  "book": {
    "id": 1,
    "title": "Machine Learning Basics",
    "author": "Alice Johnson",
    "price": 600,
    "isbn": "9781111111111",
    "status": "Available"
  }
}
```

@ Download

Response headers

```
content-length: 172
content-type: application/json
date: Sun, 20 Sep 2026 18:33:14 GMT
server: uvicorn
```

Responses

NameDescription

book_id * required

integer
(path)

103

Request body required

application/json

Edit Value Schema

"title": "Advanced Machine Learning",
"author": "Alice Johnson",
"price": 759,
.. ..: "9781111111111",
"status": "Available"

Execute

Clear

Responses

200 OK

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/books' \
-H 'accept: */*'
```



Request URL

```
http://127.0.0.1:8000/books
```

Server response

Code	Details
200	Response body

```
{
  "id": 1,
  "title": "Python Programming",
  "author": "John Smith",
  "price": 10,
  "isbn": "9781234567898",
  "status": "Avaihble"
},
{
  "id": 2,
  "title": "Database Management",
  "author": "Robert Brown",
  "price": 15,
  "isbn": "9789876543210",
  "status": "Av.1ihble"
}
```



Download

Response headers

```
content-length: 240
content-type: application/json
date: Sun, 20 Sep 2026 18:30:48 GMT
server: uvicorn
```

Responses

Code	Description	Links
------	-------------	-------

Responses

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/admin/books/103' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Advanced Machine Learning",
    "author": "Alice Johnson",
    "price": 750,
    "isbn": "9781111111111",
    "status": "Available"
  }'
```

Request URL

http://127.0.0.1: 8000/admin/books/103

Server response

Code Details

200

Response body

```
{
  "message": "Book updated successfully.",
  "book": {
    "id": 103,
    "title": "Advanced Machine Learning",
    "author": "Alice Johnson",
    "price": 750,
    "isbn": "9781111111111",
    "status": "Available"
  }
}
```

 Download

Response headers

```
content-length: 176
content-type: application/Json
date: Sun, 20 Sep 2026 18:34:42 GMT
server: uvicorn
```

Responses